

CAMERA – LIDAR EXTRINSIC CALIBRATION ON THE FLY

Presented by:

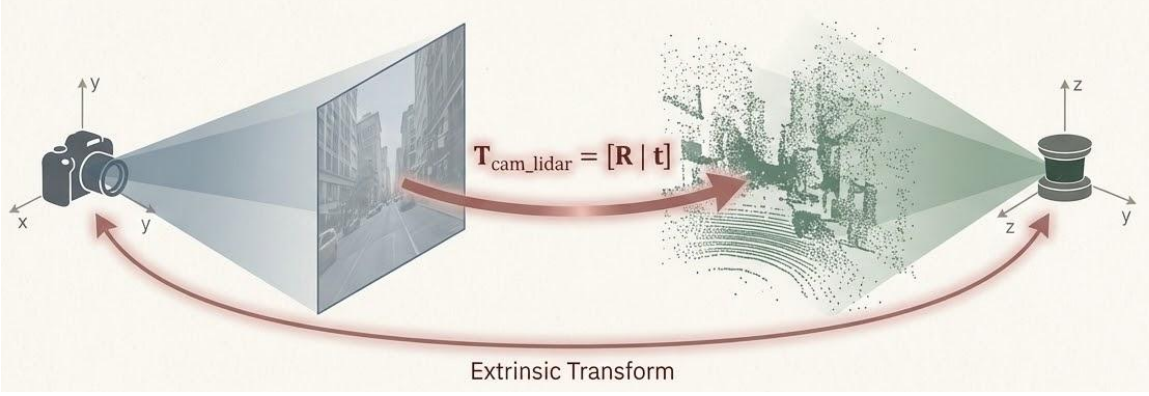
Hamza Naeem
Gowtham Parasuraman
Sairam Sridharan
Ronit Shetty

Northeastern University

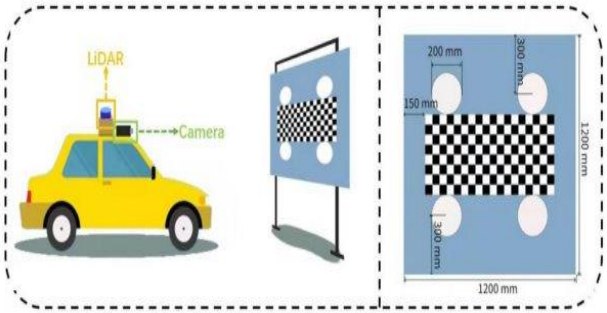
December 8, 2025

Why Camera–LiDAR Calibration Is Essential?

- Each sensor sees the world differently:
  - Camera: Rich texture, color, edges, semantics
  - LiDAR: 3D geometry, depth, range discontinuities
- To interpret objects correctly, a robot must fuse these two modalities
- Even small calibration errors cause:
  - Incorrect projection of LiDAR points onto the image
  - Poor perception (misaligned obstacles, incorrect depth)
  - Faulty object detection and tracking
  - Degraded SLAM, mapping, and navigation performance



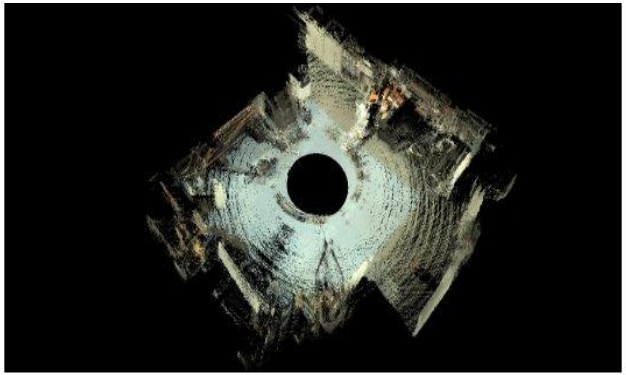
Extrinsic Calibration Approaches



Target Based Methods:

Use known structured patterns (checkerboards, AprilTags, polygon boards).

Pros: High accuracy
Cons: Need controlled environment, physical targets, not scalable outdoors



Targetless Methods:

Exploit natural features in the environment (edges, planes, intensity gradients).

Pros: No calibration board required
Cons: Sensitive to scene structure; may require good motion

Why Targetless & Online Calibration?

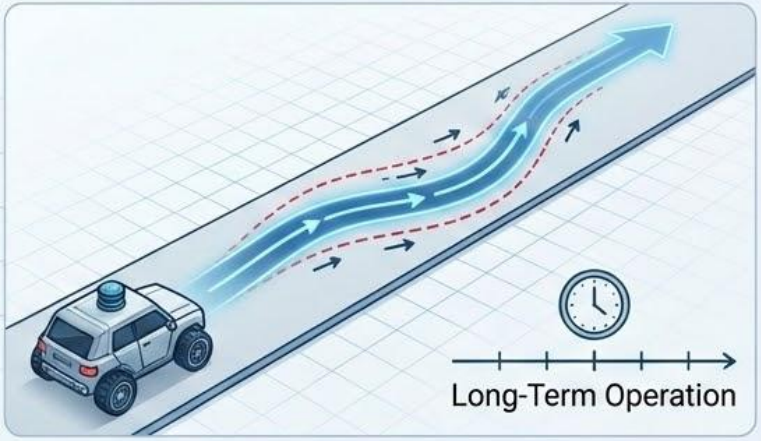
Targetless Calibration: Exploiting **Natural Features**



No artificial targets needed; uses natural geometric cues like edges and planes from real-world scenes.

- **Fuses** 3D point cloud and 2D image edges.
- **Calibrates** directly from environment.
- **Eliminates** need for calibration boards.

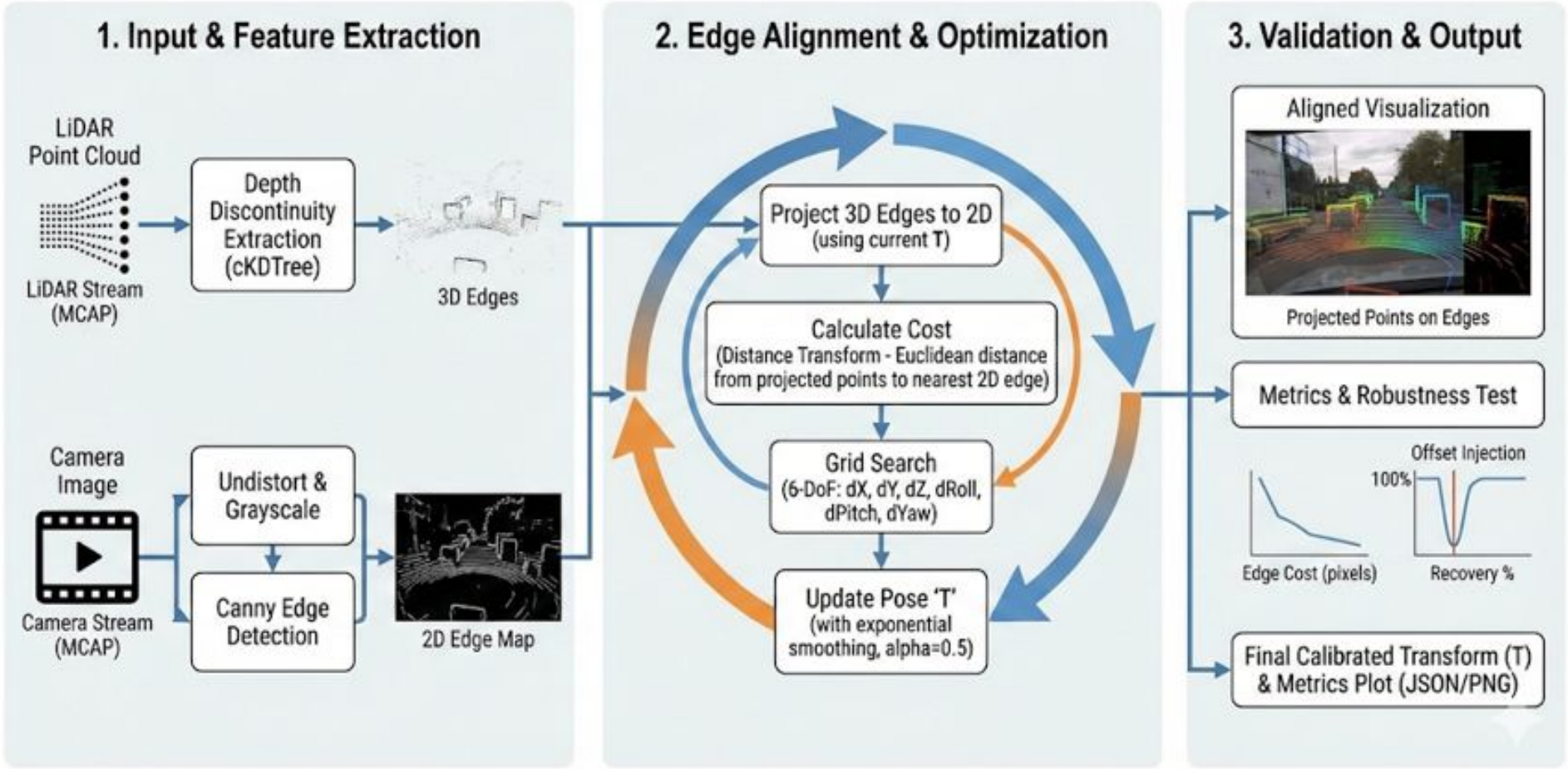
Online Calibration: **Continuous** & Adaptive



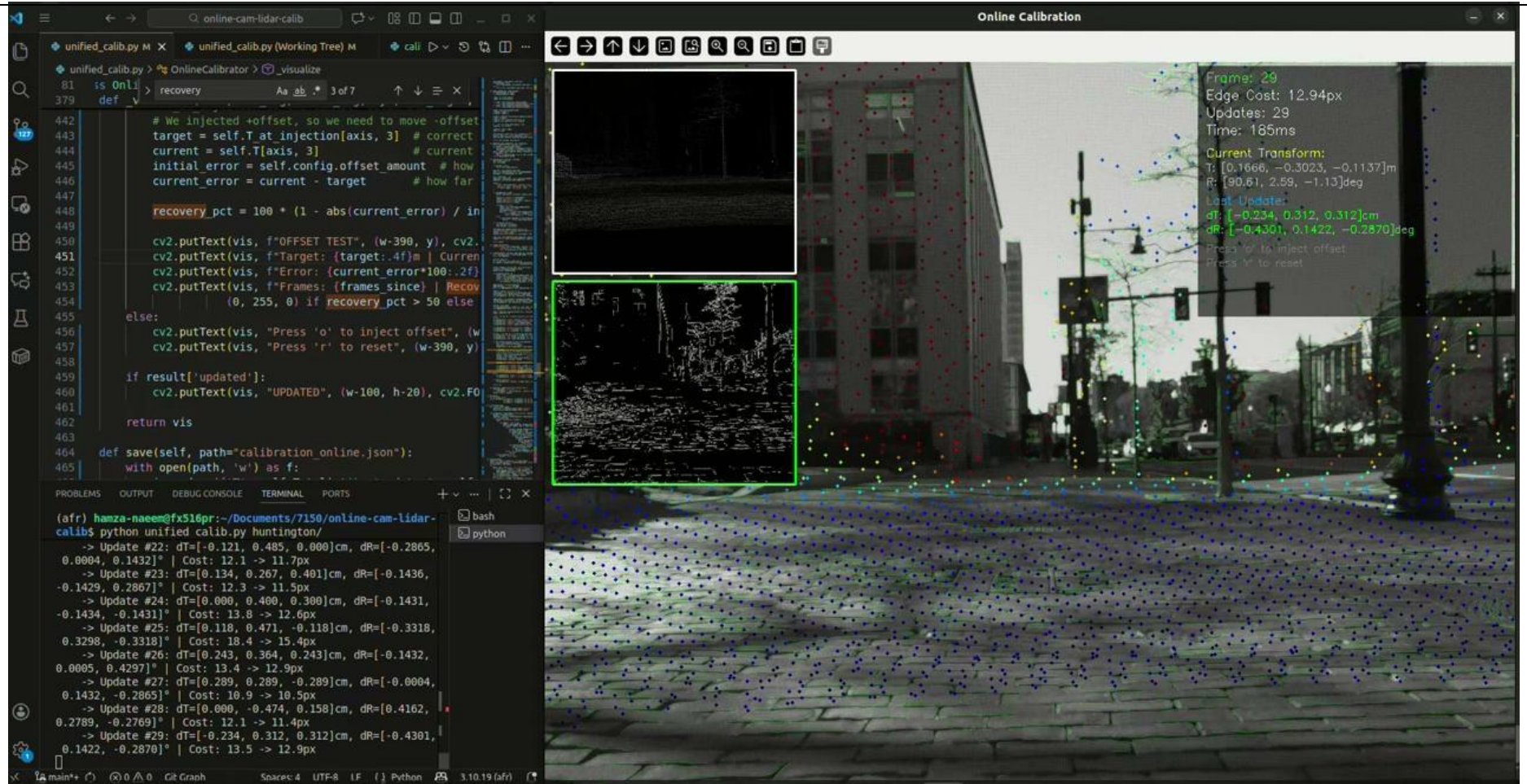
Continuously estimates and updates extrinsic transform to counter mechanical drift and changes.

- Addresses drift from vibration & temperature.
- Applies gentle, **incremental** adjustments.
- Ensures long-term accuracy and stability.

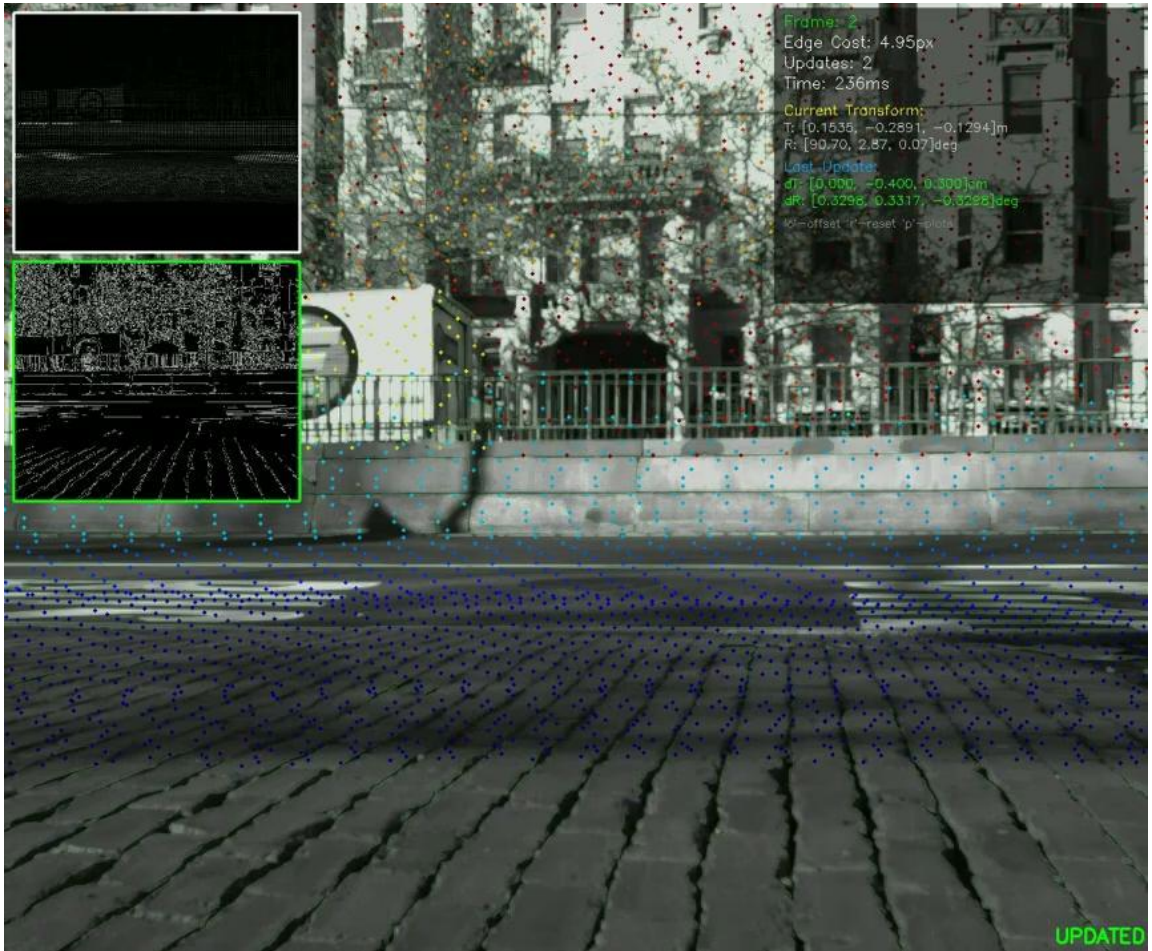
Methodology



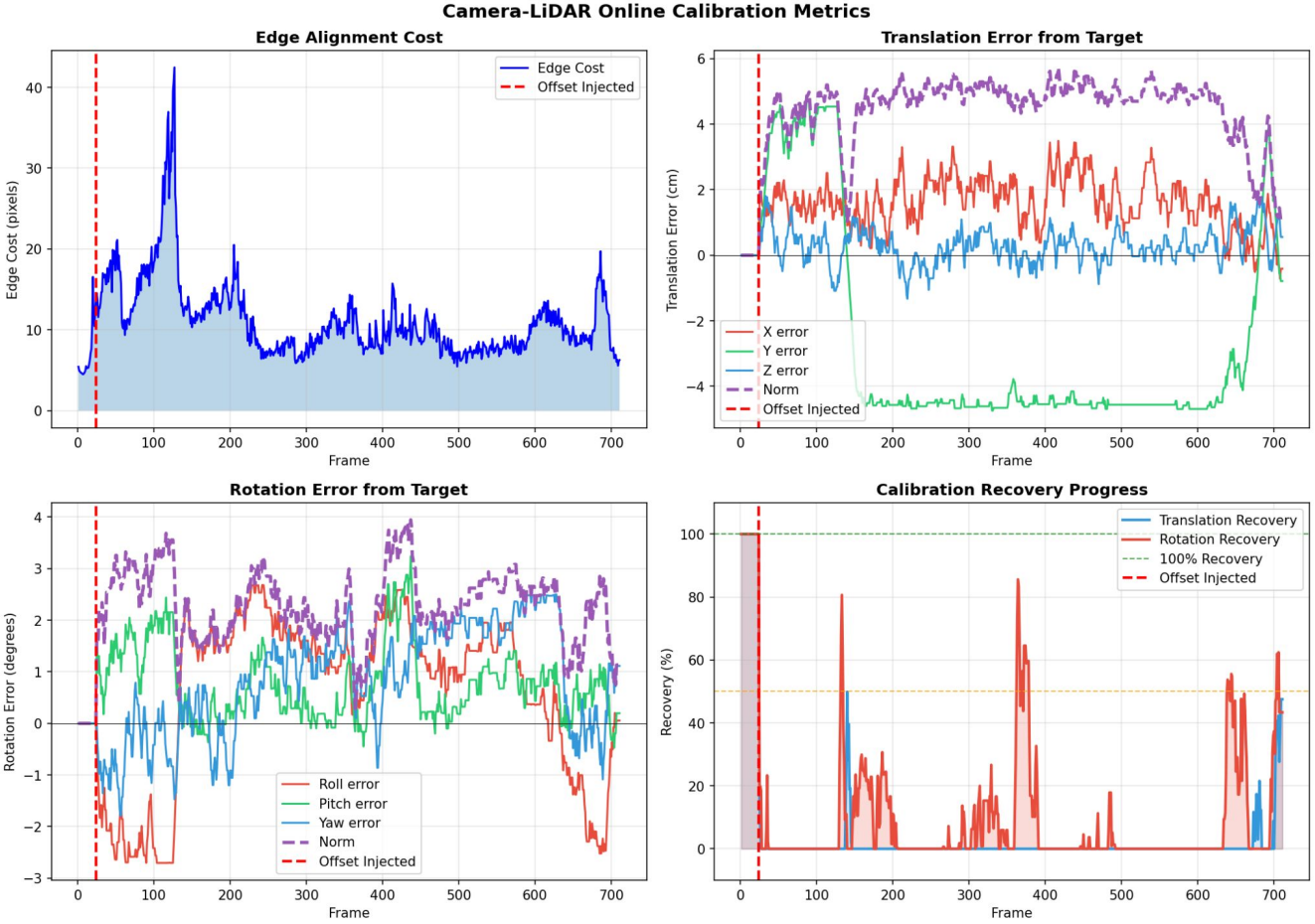
NeURoam Dataset (Huntington) - Offset (x = 2 cm)



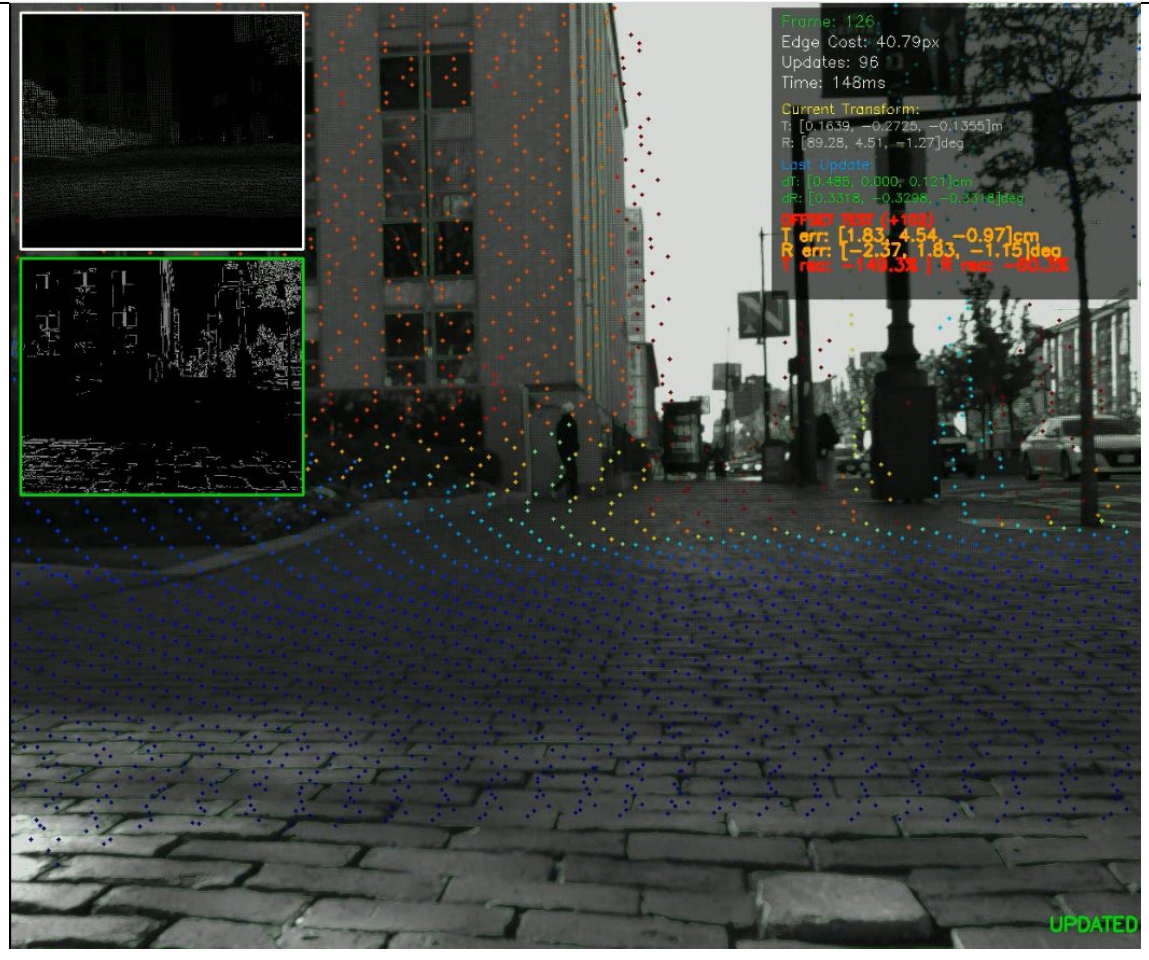
NeURoam Dataset (Huntington) - Offset (R + t)



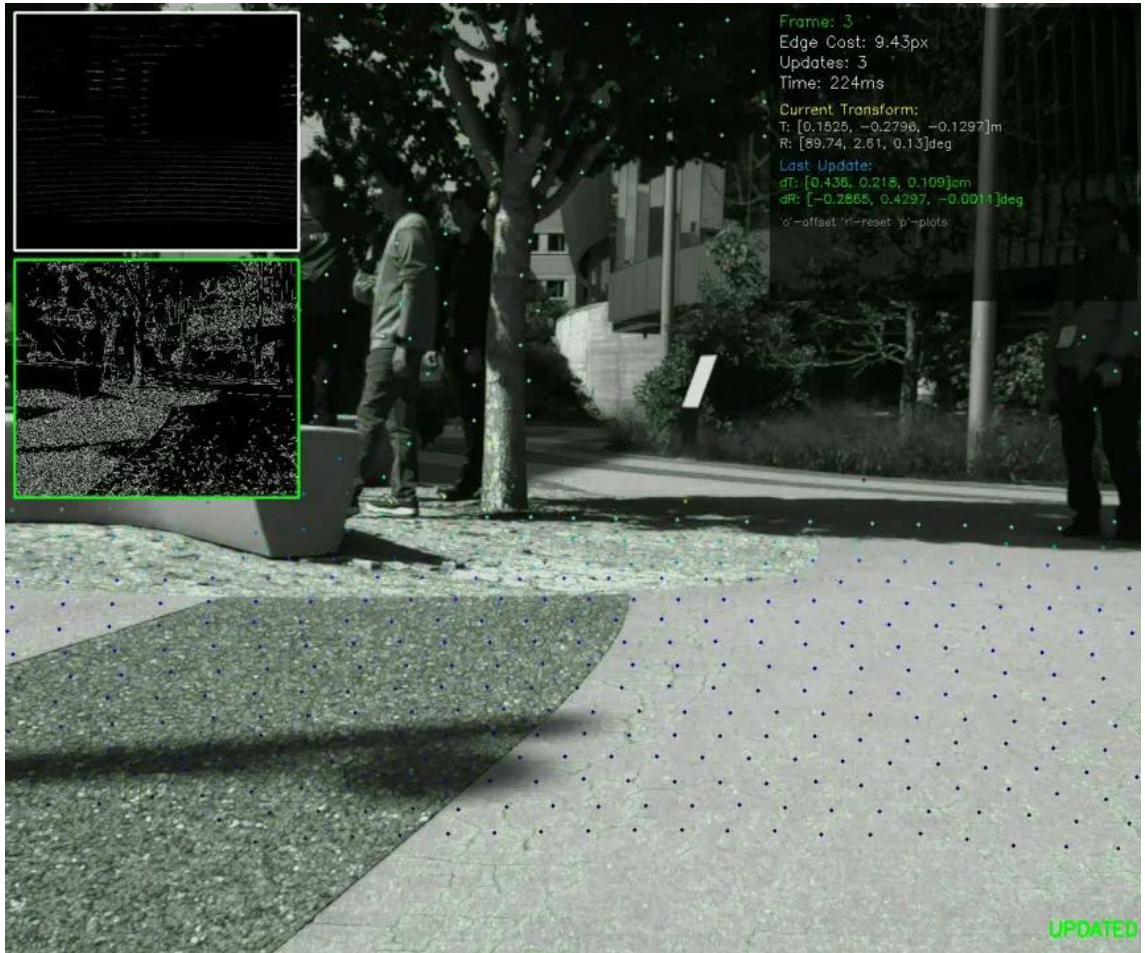
NeURoam Dataset (Huntington) - Offset (R + t)



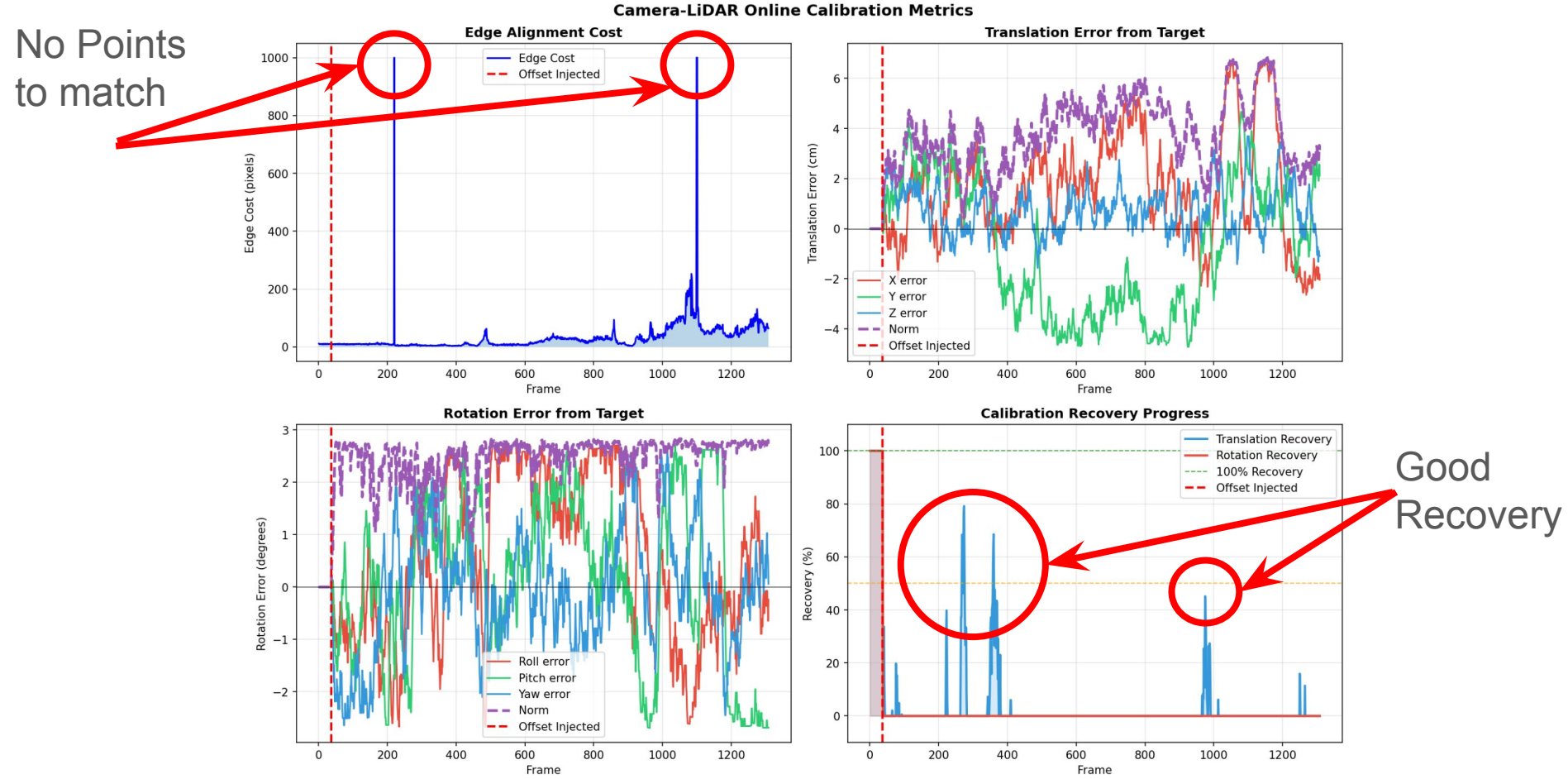
NeURoam Dataset (Huntington) - Offset (R + t)



NeURoam Dataset (ISEC Bridge) - Offset (x = 2 cm)



NeURoam Dataset (ISEC Bridge) - Offset (x = 2 cm)



Peaks at Frame 220 and 1099

-> Drift Penalty - Not Enough Points - 1000

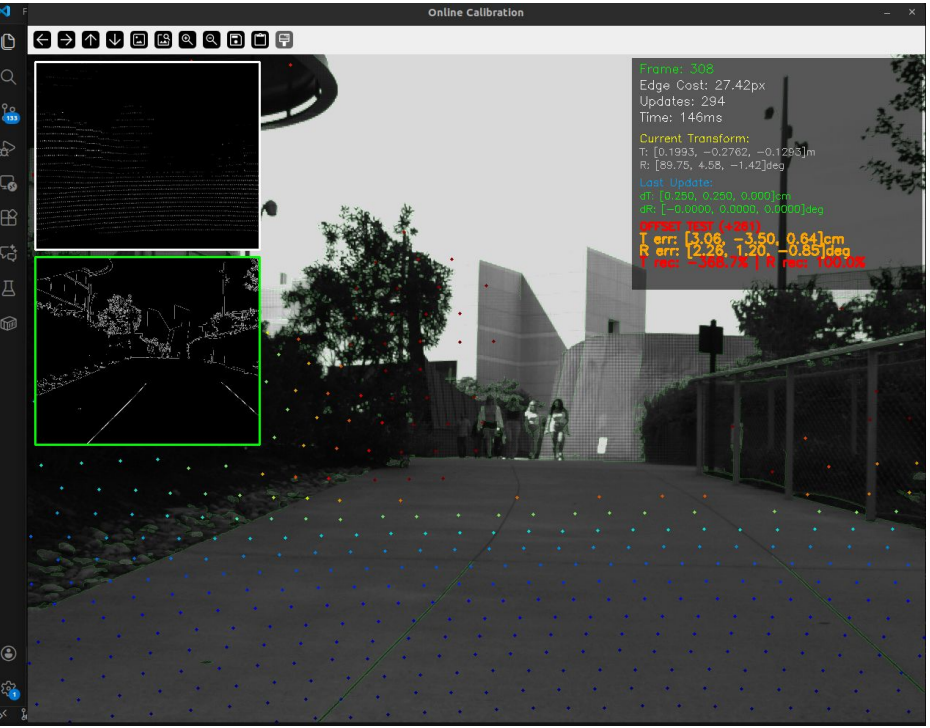




Limitations & Future Work

1. We optimize Translation first, locks it, and then optimizes Rotation - No Coupling affect taken into account.
   a. Joint Optimization can fix that.
2. The "Grid Search" Trap - Limited Step Size
   a. LM or better optimization ??
   b. Local Minima
3. Better Outlier Rejection - Instead of hardcoded high costs
4. Environment Limitations
   a. Flat Environment - No Edges for the Lidar
   b. Texture Less Environment - No Edges for the Camera

Textureless / Flat Surfaces





NeURoam Dataset (Huntington) - SAM

